

UCI EDA

: 2026-01-07T22:42:49.723817

  uci   CSV   , , .

: category_tree

: ./uci/category_tree.csv

: (1669, 2)

 5 

head()

```
categoryid parentid 0 1016 213.0 1 809 169.0 2 570 9.0 3 1691 885.0 4 536 1691.0 `` ###   (info) ## info()
```

RangeIndex: 1669 entries, 0 to 1668 Data columns (total 2 columns): # Column Non-Null Count Dtype

--- -----

0 categoryid 1669 non-null int64

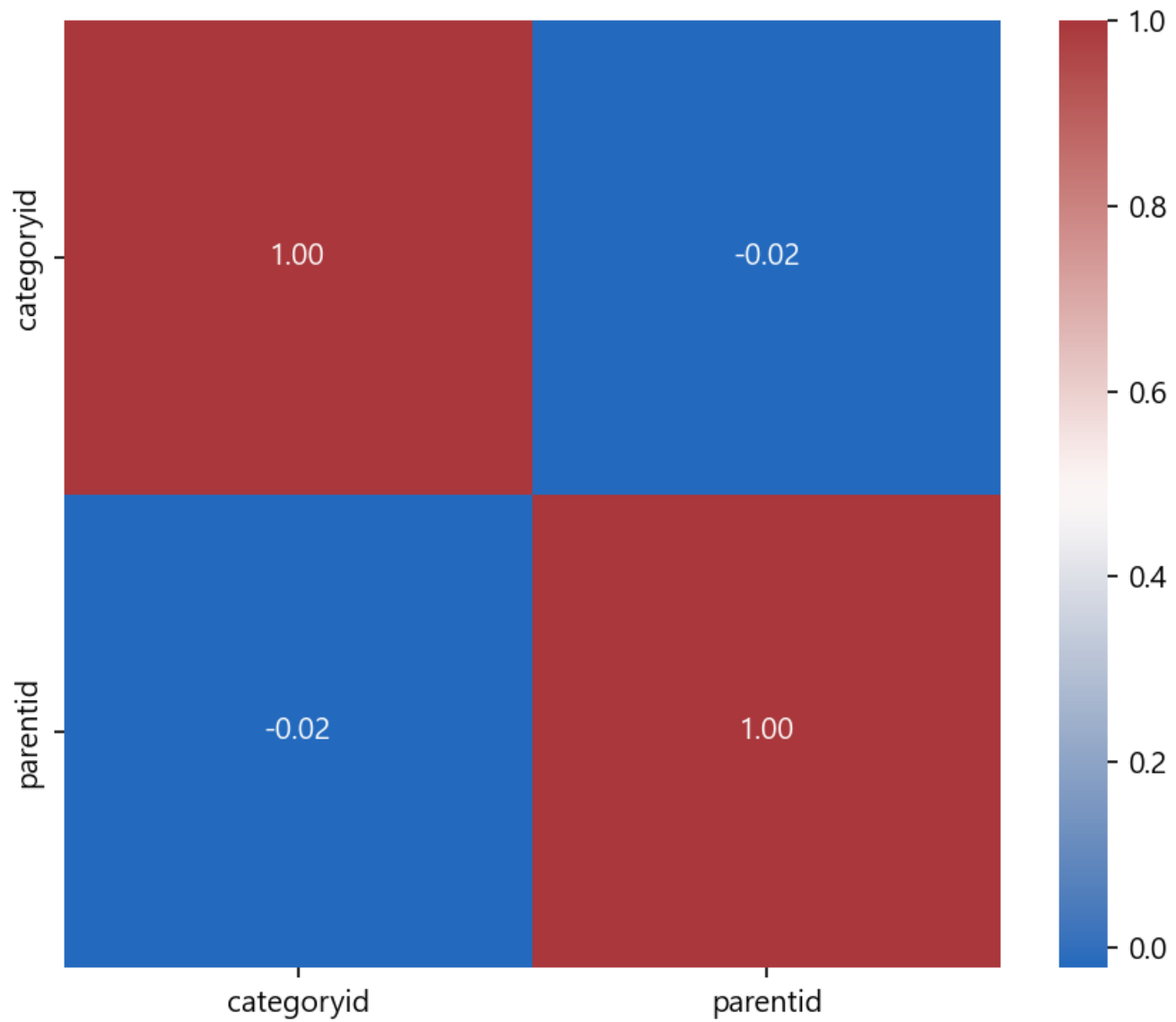
1 parentid 1644 non-null float64 dtypes: float64(1), int64(1) memory usage: 26.2 KB ``

 (describe())

describe()

```
categoryid parentid count 1669.000000 1644.000000 mean 849.285201 847.571168 std 490.195116 505.058485 min 0.000000
8.000000 25% 427.000000 381.000000 50% 848.000000 866.000000 75% 1273.000000 1291.000000 max 1698.000000 1698.000000 ``
###   ## missing values (count)
```

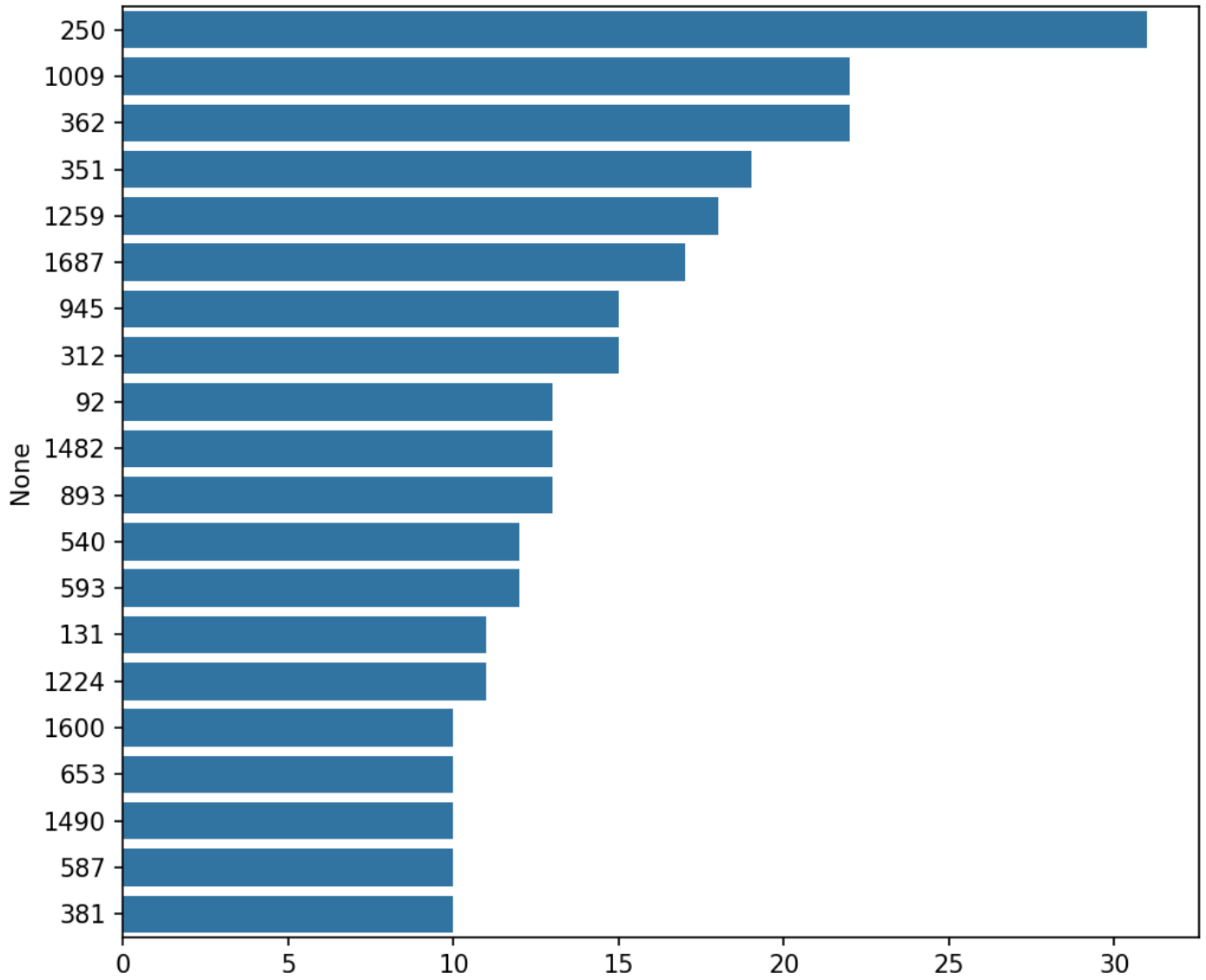
```
parentid 25 categoryid 0 ``
```



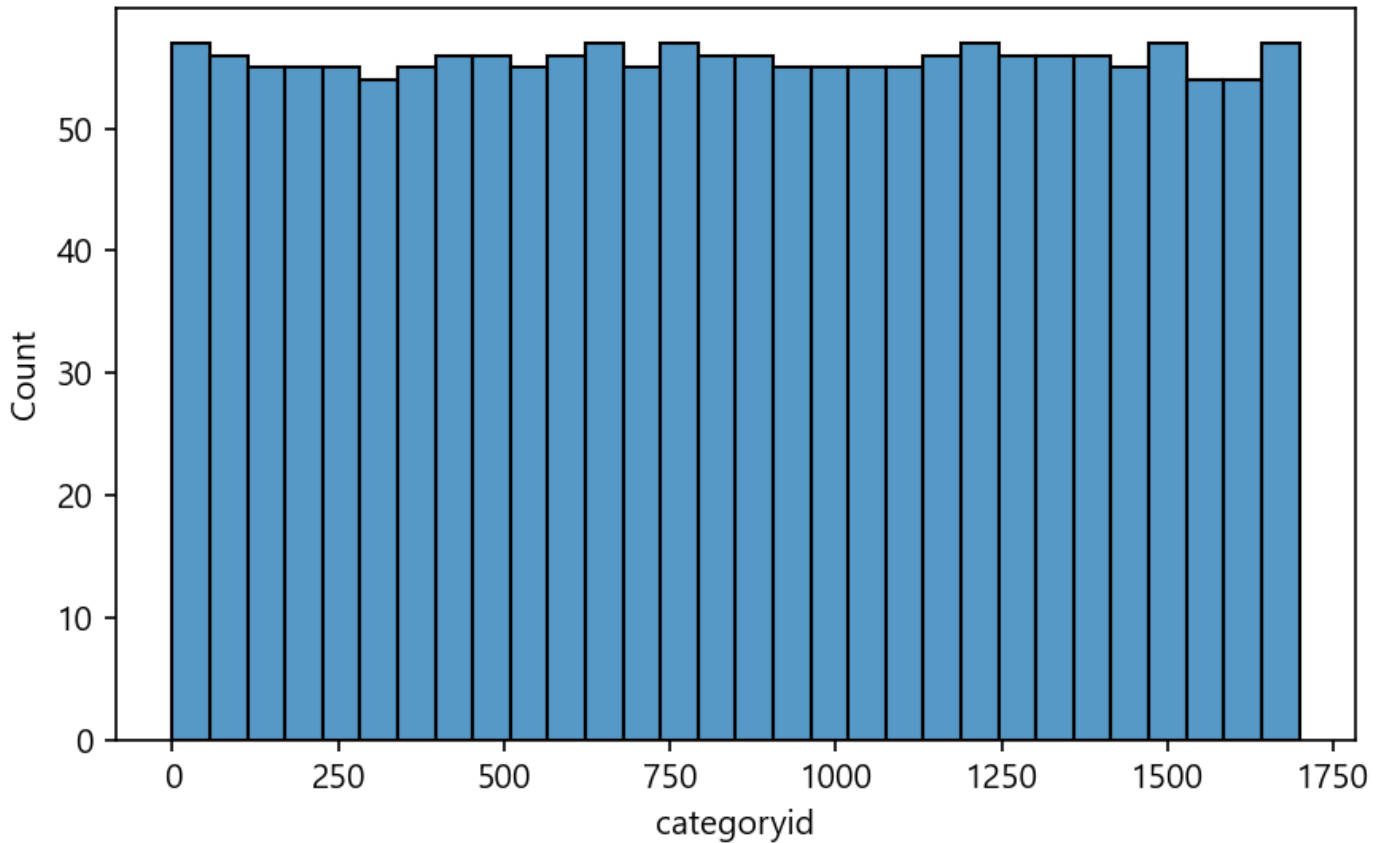
category depth counts

```
0 25 1 174 2 702 3 665 4 90 5 13 `` ![depth](./images/category_tree_depth_dist.png) ## top parents by child count
250 31 1009 22 362 22 351 19 1259 18 1687 17 945 15 312 15 92 13 1482 13 893 13 540 12 593 12 131 11 1224 11 1600 10 653 10 1490 10 587 10 381 10 ``
```

Top parents by child count



category_tree - categoryid 분포



categoryid ■■■■

```
count 1669.000000 mean 849.285201 std 490.195116 min 0.000000 25% 427.000000 50% 848.000000 75% 1273.000000 max 1698.000000 `` ![hist](./images/category_tree_hist_parentid.png) ## parentid ■■■■
```

```
count 1644.000000 mean 847.571168 std 505.058485 min 8.000000 25% 381.000000 50% 866.000000 75% 1291.000000 max 1698.000000 ``
```

■■■ ■■

- ■■■■ category_tree ■(■) ■ 1669■■■ ■■ 2■■■ ■■ ■■■■.
- ■■■■ ■■ ■■ ■■■■ ■■■■(■■/■■)■ ■■■■.
- ■■ ■■ ■ ■■ ■■ ■■■■ ■■ ■■■■ ■■■■.

■■■■: events

■■■■: ./uci/events.csv

■■: (2756101, 5)

■■ 5 ■ ■

head()

```
timestamp visitorid event itemid transactionid 0 1433221332117 257597 view 355908 NaN 1 1433224214164 992329 view
248676 NaN 2 1433221999827 111016 view 318965 NaN 3 1433221955914 483717 view 253185 NaN 4 1433221337106 951259 view
367447 NaN `` ### ■■■■ ■■ (info) ## info()
```

RangeIndex: 2756101 entries, 0 to 2756100 Data columns (total 5 columns): # Column Dtype

0 timestamp int64

1 visitorid int64

2 event object 3 itemid int64

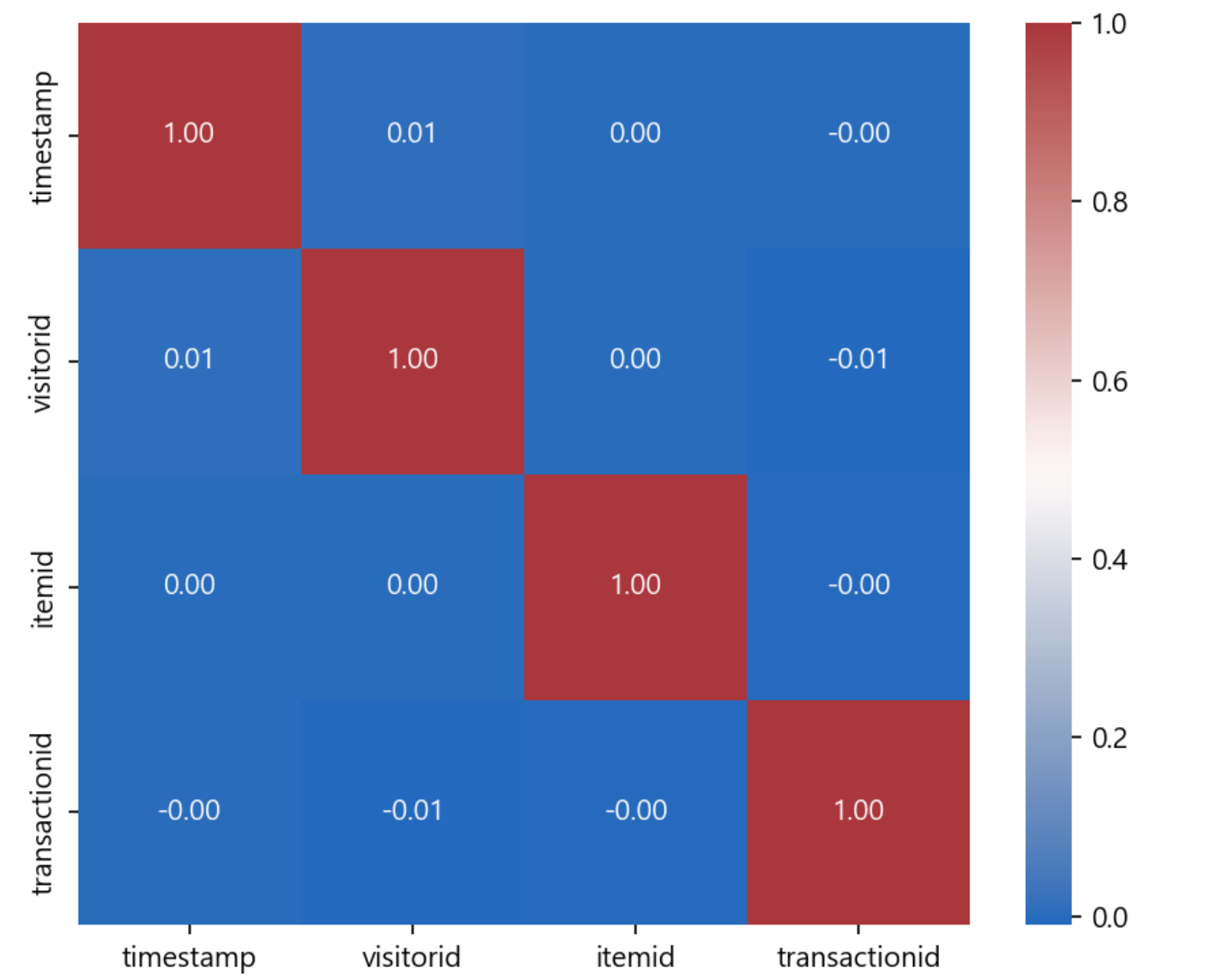
4 transactionid float64 dtypes: float64(1), int64(3), object(1) memory usage: 105.1+ MB ``

■■ ■■ (describe())

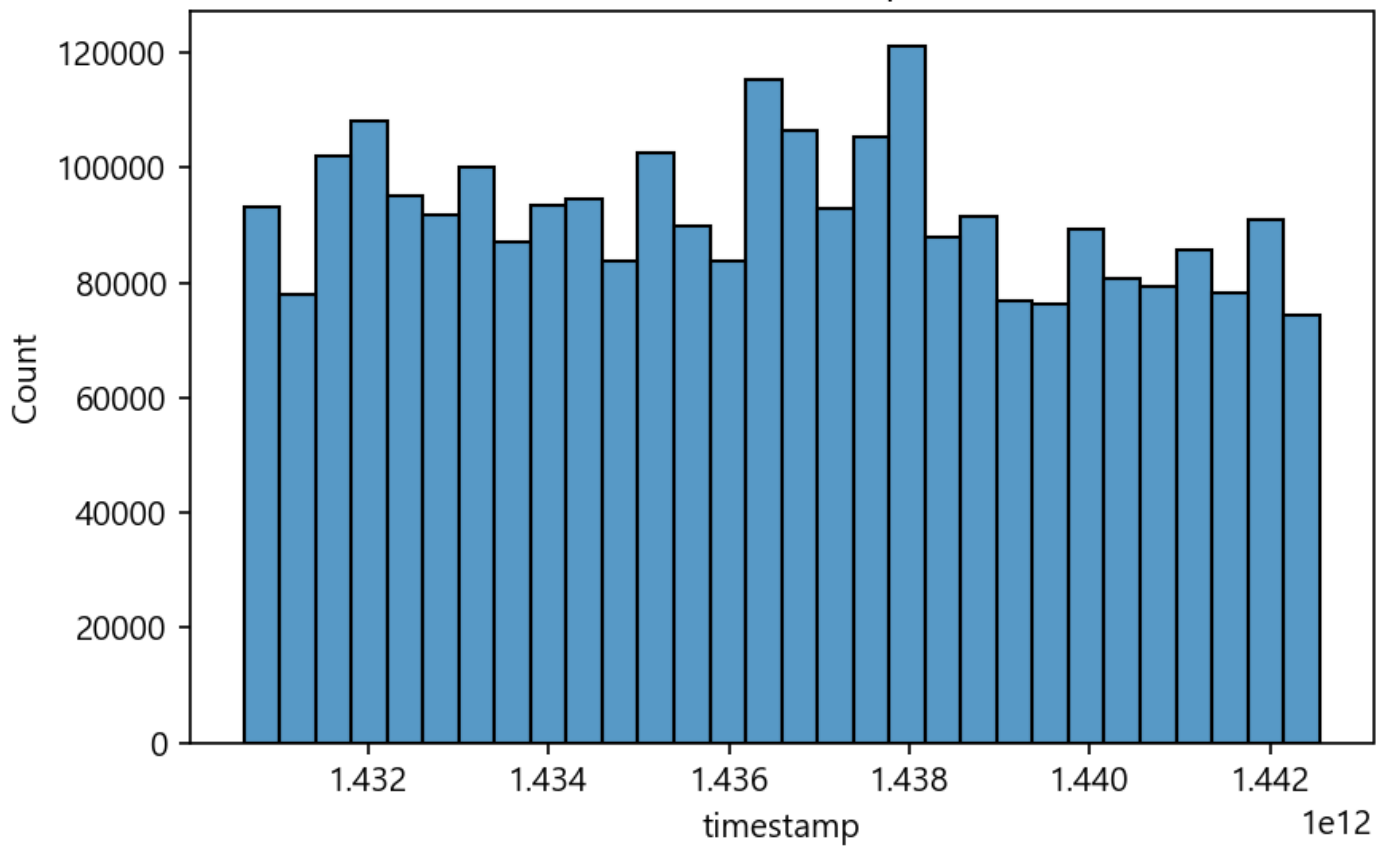
describe()

timestamp visitorid event itemid transactionid count 2.756101e+06 2.756101e+06 2756101 2.756101e+06 22457.000000 unique
NaN NaN 3 NaN NaN top NaN NaN view NaN NaN freq NaN NaN 2664312 NaN NaN mean 1.436424e+12 7.019229e+05 NaN 2.349225e+05
8826.497796 std 3.366312e+09 4.056875e+05 NaN 1.341954e+05 5098.996290 min 1.430622e+12 0.000000e+00 NaN 3.000000e+00
0.000000 25% 1.433478e+12 3.505660e+05 NaN 1.181200e+05 4411.000000 50% 1.436453e+12 7.020600e+05 NaN 2.360670e+05
8813.000000 75% 1.439225e+12 1.053437e+06 NaN 3.507150e+05 13224.000000 max 1.442545e+12 1.407579e+06 NaN 4.668670e+05
17671.000000 `` ### ████ ███ ## missing values (count)

transactionid 2733644 timestamp 0 visitorid 0 event 0 itemid 0 ``



events - timestamp 분포

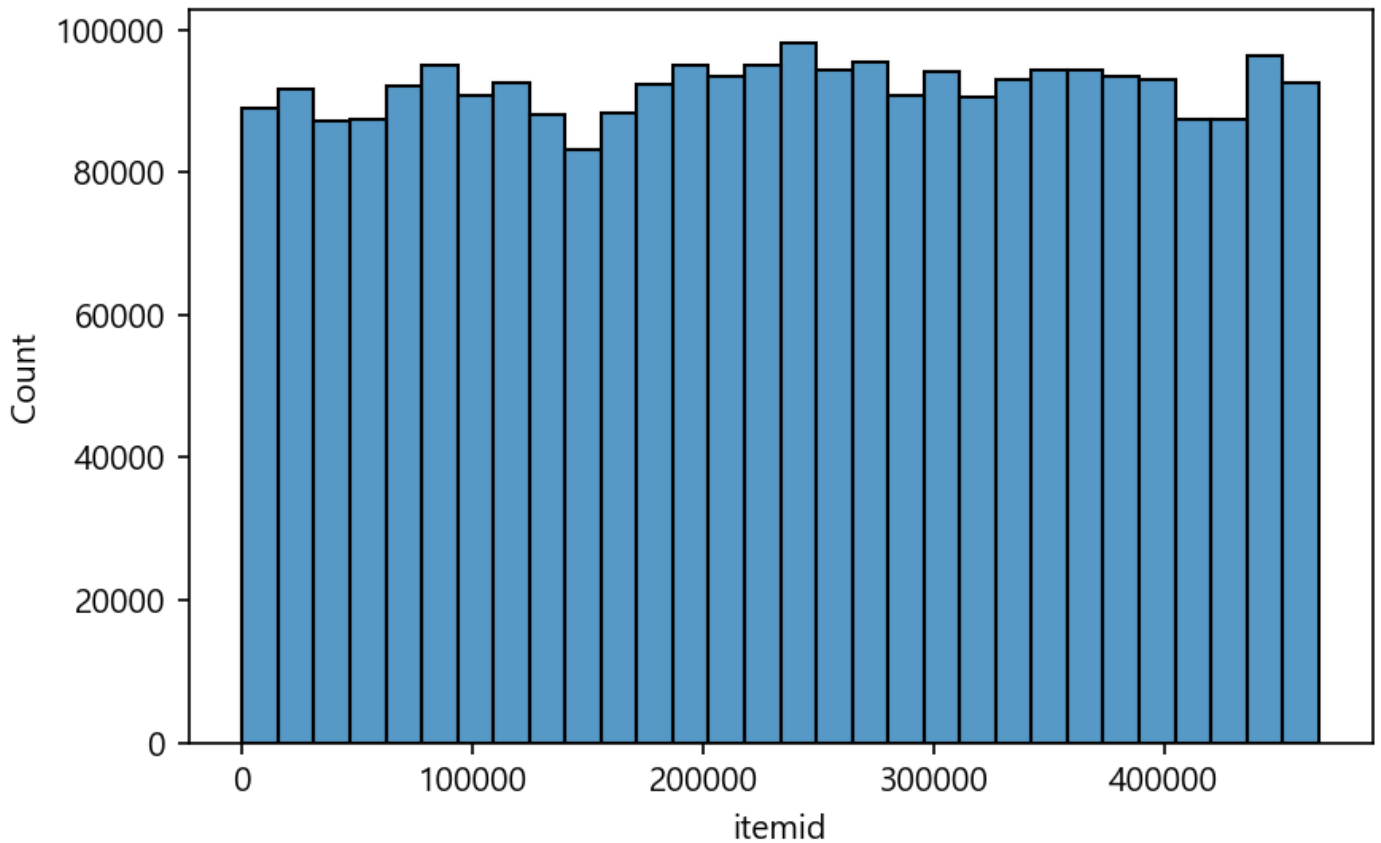


timestamp ■■■■

```
count 2.756101e+06 mean 1.436424e+12 std 3.366312e+09 min 1.430622e+12 25% 1.433478e+12 50% 1.436453e+12 75%
1.439225e+12 max 1.442545e+12 `` ![hist](./images/events_hist_visitorid.png) ## visitorid ■■■■
```

```
count 2.756101e+06 mean 7.019229e+05 std 4.056875e+05 min 0.000000e+00 25% 3.505660e+05 50% 7.020600e+05 75% 1.053437e+06 max 1.407579e+06
``
```

events - itemid 분포



itemid ■■■■

```
count 2.756101e+06 mean 2.349225e+05 std 1.341954e+05 min 3.000000e+00 25% 1.181200e+05 50% 2.360670e+05 75%
3.507150e+05 max 4.668670e+05 `` ![bar](./images/events_bar_event.png) ##### ■■ ■■■■■■ ■■ ■■ (■■ 20) ## event
value_counts
```

```
event view 2664312 addtocart 69332 transaction 22457 ``
```

event vs timestamp (count/mean)

```
count mean event view 2664312 1.436422e+12 addtocart 69332 1.436514e+12 transaction 22457 1.436469e+12 ``
![box](./images/events_box_event_timestamp.png) ### ■■■■ ■■ - ■■■■ `events` ■(■) ■ 2756101■■ ■■ 5■■ ■■ ■■■■. -
■■■■ ■■ ■■ ■■ ■■■(■■/■■)■ ■■■■■. - ■■ ■■ ■ ■■ ■■ ■■■■ ■■ ■■■■ ■■■■. --- ## ■■■■: item_properties_part1
■■■■: ./uci/item_properties_part1.csv ■■: (10999999, 4) ### ■■ 5■ ■ ## head()
```

```
timestamp itemid property value
```

```
0 1435460400000 460429 categoryid 1338 1 1441508400000 206783 888 1116713 960601 n277.200 2 1439089200000 395014 400 n552.000 639502
n720.000 424566 3 1431226800000 59481 790 n15360.000 4 1431831600000 156781 917 828513 ``
```

■■■■ ■■ (info)

info()

```
<class 'pandas.core.frame.DataFrame'> RangeIndex: 10999999 entries, 0 to 10999998 Data columns (total 4 columns): #
Column Dtype --- ----- 0 timestamp int64 1 itemid int64 2 property object 3 value object dtypes: int64(2),
object(2) memory usage: 335.7+ MB `` ### ■■ ■■ (`describe()`) ## describe()
```

```
timestamp itemid property value
```

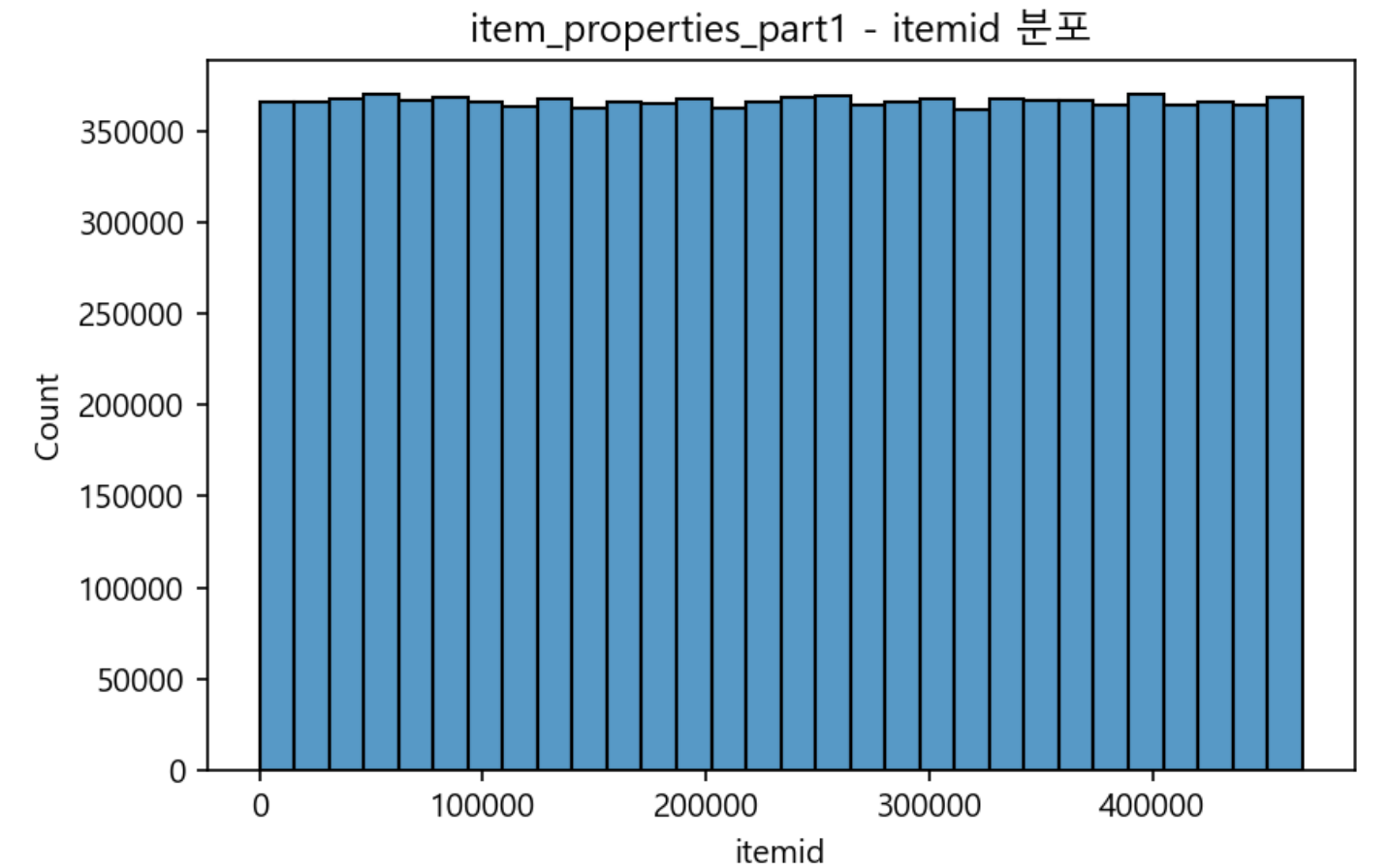
```
count 1.100000e+07 1.100000e+07 10999999 10999999 unique NaN NaN 1097 1231581 top NaN NaN 888 769062 freq NaN NaN 1629817 833710 mean
1.435158e+12 2.333851e+05 NaN NaN std 3.327653e+09 1.348258e+05 NaN NaN min 1.431227e+12 0.000000e+00 NaN NaN 25% 1.432436e+12
1.165150e+05 NaN NaN 50% 1.433646e+12 2.334990e+05 NaN NaN 75% 1.437880e+12 3.501860e+05 NaN NaN max 1.442113e+12 4.668660e+05 NaN
NaN ``
```

■■■■ ■■

missing values (count)

timestamp 0 itemid 0 property 0 value 0 `` ![corr](./images/item_properties_part1_corr.png)
![hist](./images/item_properties_part1_hist_timestamp.png) ## timestamp ■■■■

count 1.100000e+07 mean 1.435158e+12 std 3.327653e+09 min 1.431227e+12 25% 1.432436e+12 50% 1.433646e+12 75% 1.437880e+12 max 1.442113e+12
``



itemid ■■■■

count 1.100000e+07 mean 2.333851e+05 std 1.348258e+05 min 0.000000e+00 25% 1.165150e+05 50% 2.334990e+05 75%
3.501860e+05 max 4.668660e+05 `` ![bar](./images/item_properties_part1_bar_property.png) ##### ■■ ■■■■■■ ■■ ■■ (■■
20) ## property value_counts

property 888 1629817 790 970800 available 817387 categoryid 426305 6 343207 283 323681 776 311654 678 261829 364 256340 202 242984 839 226921 159
226502 917 226437 764 226242 112 226102 227 188209 698 157281 451 142388 663 131331 962 128976 ``

property vs timestamp (count/mean)

count mean property 888 1629817 1.436446e+12 790 970800 1.436099e+12 available 817387 1.435914e+12 categoryid 426305
1.434699e+12 6 343207 1.434305e+12 283 323681 1.434004e+12 776 311654 1.433997e+12 678 261829 1.433278e+12 364 256340
1.433214e+12 202 242984 1.434872e+12 839 226921 1.434788e+12 159 226502 1.432696e+12 917 226437 1.432701e+12 764 226242
1.432692e+12 112 226102 1.432692e+12 227 188209 1.434644e+12 698 157281 1.434822e+12 451 142388 1.437148e+12 663 131331
1.437195e+12 962 128976 1.437251e+12 `` ![bar](./images/item_properties_part1_bar_value.png) ##### ■■ ■■■■■■ ■■ ■■
(■■ 20) ## value value_counts

value 769062 833710 0 469704 1 348216 519769 226502 1285872 226242 679677 226102 1116693 85354 1297729 n156.000 606827 65224 664227 63768
664227 1305534 664227 463202 664227 52404 n552.000 639502 n720.000 424566 40680 150169 610517 39789 1115994 29810 664227 1305534 664227
27589 1154859 27067 150169 274770 79212 26663 1037891 26384 n12.000 25179 1141052 n48.000 140719 553394 23466 1152934 1238769 23434 ``

value vs timestamp (count/mean)

count mean value 769062 833710 1.434739e+12 0 469704 1.435495e+12 1 348216 1.436474e+12 519769 226502 1.432696e+12
1285872 226242 1.432692e+12 679677 226102 1.432692e+12 1116693 85354 1.436290e+12 1297729 n156.000 606827 65224
1.439375e+12 664227 63768 1.434559e+12 664227 1305534 664227 463202 664227 52404 1.439446e+12 n552.000 639502 n720.000
424566 40680 1.438490e+12 150169 610517 39789 1.434678e+12 1115994 29810 1.435879e+12 664227 1305534 664227 27589


```
1.439468e+12 1154859 27067 1.434960e+12 150169 274770 79212 26663 1.434755e+12 1037891 26384 1.434114e+12 n12.000 25179
1.434381e+12 1141052 n48.000 140719 553394 23466 1.439594e+12 1152934 1238769 23434 1.435882e+12 ``
! [box](./images/item_properties_part1_box_property_timestamp.png) ###  -  `item_properties_part1`  ( ) 
10999999  4  . -  ( / )  . -  5  ## head()
## : item_properties_part2 : ./uci/item_properties_part2.csv : (9275903, 4) ### 5  ## head()

timestamp itemid property value

0 1433041200000 183478 561 769062 1 1439694000000 132256 976 n26.400 1135780 2 1435460400000 420307 921 1149317 1257525 3 1431831600000
403324 917 1204143 4 1435460400000 230701 521 769062 ``
```

(info)

info()

```
<class 'pandas.core.frame.DataFrame'> RangeIndex: 9275903 entries, 0 to 9275902 Data columns (total 4 columns): #
Column Dtype --- ----- 0 timestamp int64 1 itemid int64 2 property object 3 value object dtypes: int64(2),
object(2) memory usage: 283.1+ MB `` ###  ( `describe()` ) ## describe()

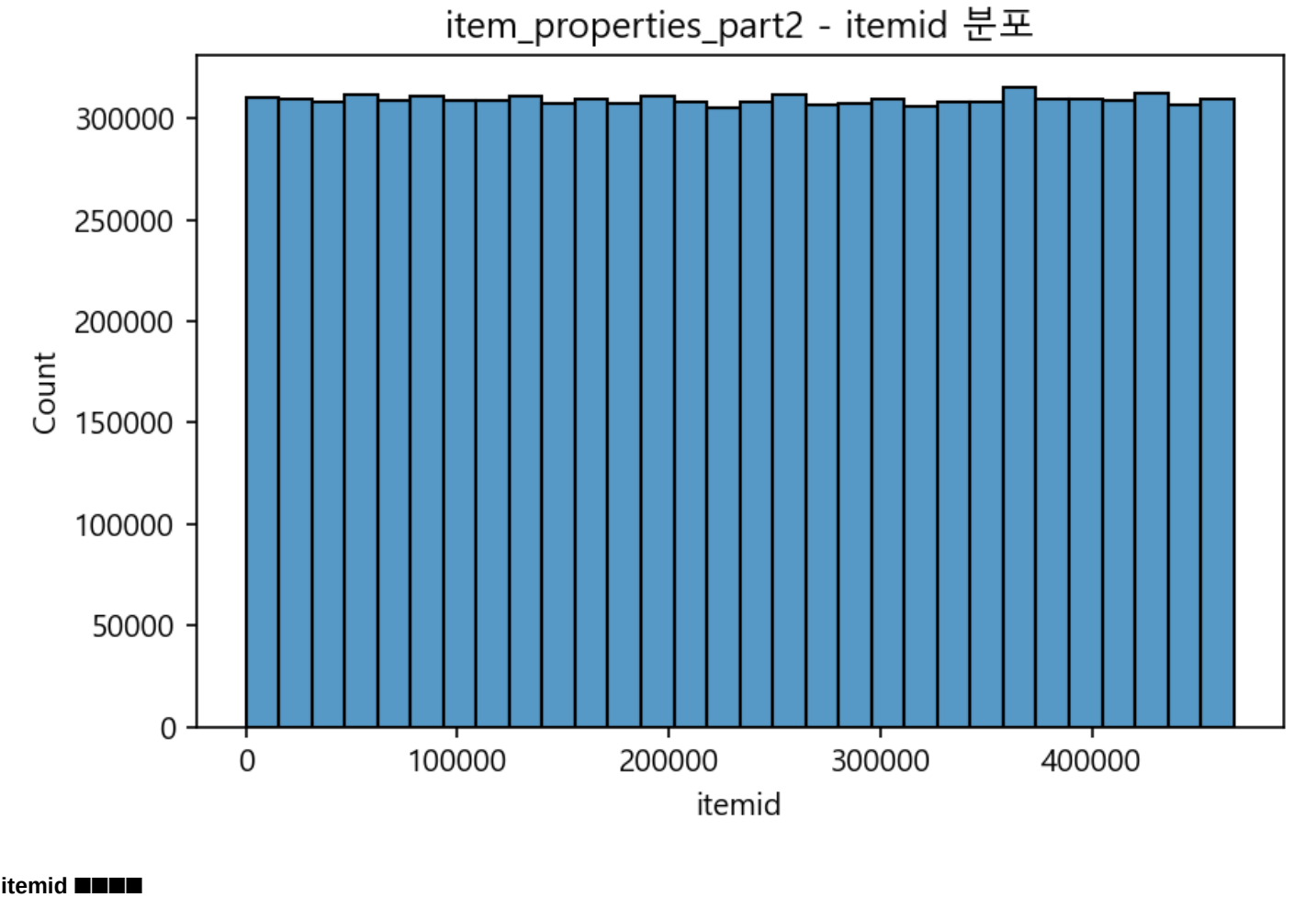
timestamp itemid property value

count 9.275903e+06 9.275903e+06 9275903 9275903 unique NaN NaN 1094 1075730 top NaN NaN 888 769062 freq NaN NaN 1370581 703537 mean
1.435156e+12 2.333968e+05 NaN NaN std 3.327970e+09 1.348682e+05 NaN NaN min 1.431227e+12 0.000000e+00 NaN NaN 25% 1.432436e+12
1.165175e+05 NaN NaN 50% 1.433646e+12 2.334620e+05 NaN NaN 75% 1.437880e+12 3.504470e+05 NaN NaN max 1.442113e+12 4.668660e+05 NaN
NaN ``
```

missing values (count)

```
timestamp 0 itemid 0 property 0 value 0 `` ! [corr](./images/item_properties_part2_corr.png)
! [hist](./images/item_properties_part2_hist_timestamp.png) ## timestamp 

count 9.275903e+06 mean 1.435156e+12 std 3.327970e+09 min 1.431227e+12 25% 1.432436e+12 50% 1.433646e+12 75% 1.437880e+12 max 1.442113e+12
``
```



count 9.275903e+06 mean 2.333968e+05 std 1.348682e+05 min 0.000000e+00 25% 1.165175e+05 50% 2.334620e+05 75% 3.504470e+05 max 4.668660e+05 `` ![bar](./images/item_properties_part2_bar_property.png) ##### █ ████ █ █ (█ 20) ## property value_counts

property 888 1370581 790 819716 available 686252 categoryid 361909 6 288264 283 273738 776 262566 364 220146 678 220137 202 205954 112 190951 764 190811 917 190790 159 190551 839 190318 227 159283 698 132568 451 122028 962 110396 663 109482 ``

property vs timestamp (count/mean)

count mean property 888 1370581 1.436450e+12 790 819716 1.436104e+12 available 686252 1.435907e+12 categoryid 361909 1.434706e+12 6 288264 1.434272e+12 283 273738 1.434012e+12 776 262566 1.433996e+12 364 220146 1.433260e+12 678 220137 1.433277e+12 202 205954 1.434862e+12 112 190951 1.432690e+12 764 190811 1.432687e+12 917 190790 1.432697e+12 159 190551 1.432681e+12 839 190318 1.434773e+12 227 159283 1.434634e+12 698 132568 1.434808e+12 451 122028 1.437178e+12 962 110396 1.437232e+12 663 109482 1.437177e+12 `` ![bar](./images/item_properties_part2_bar_value.png) ##### █ ████ █ █ (█ 20) ## value value_counts

value 769062 703537 0 393542 1 293204 679677 190952 1285872 190811 519769 190551 1116693 71889 1297729 n156.000 606827 54912 664227 54885 664227 1305534 664227 463202 664227 43841 150169 610517 33558 n552.000 639502 n720.000 424566 33139 1115994 25158 664227 1305534 664227 24132 1154859 22908 150169 274770 79212 22413 1037891 22159 n12.000 20796 726612 20003 1141052 n48.000 140719 553394 19852 ``

value vs timestamp (count/mean)

count mean value 769062 703537 1.434739e+12 0 393542 1.435497e+12 1 293204 1.436452e+12 679677 190952 1.432690e+12 1285872 190811 1.432687e+12 519769 190551 1.432681e+12 1116693 71889 1.436316e+12 1297729 n156.000 606827 54912 1.439366e+12 664227 54885 1.434563e+12 664227 1305534 664227 463202 664227 43841 1.439442e+12 150169 610517 33558 1.434707e+12 n552.000 639502 n720.000 424566 33139 1.438471e+12 1115994 25158 1.435895e+12 664227 1305534 664227 24132 1.439454e+12 1154859 22908 1.434971e+12 150169 274770 79212 22413 1.434755e+12 1037891 22159 1.434039e+12 n12.000 20796 1.434375e+12 726612 20003 1.434276e+12 1141052 n48.000 140719 553394 19852 1.439616e+12 ``

![box](./images/item_properties_part2_box_property_timestamp.png) ### ███ █ - ████ `item\_properties\_part2` █(█) █ 9275903██ █ 4██ █ ████. - ████ █ █ ███ ███(██/██)█ ██████. - █ █ █ █ █ ███ █ ███ ██████. --- # ███ ███(████/████) █ █ █ ## ███ █ █ █ ## customers describe

transactions items_purchased

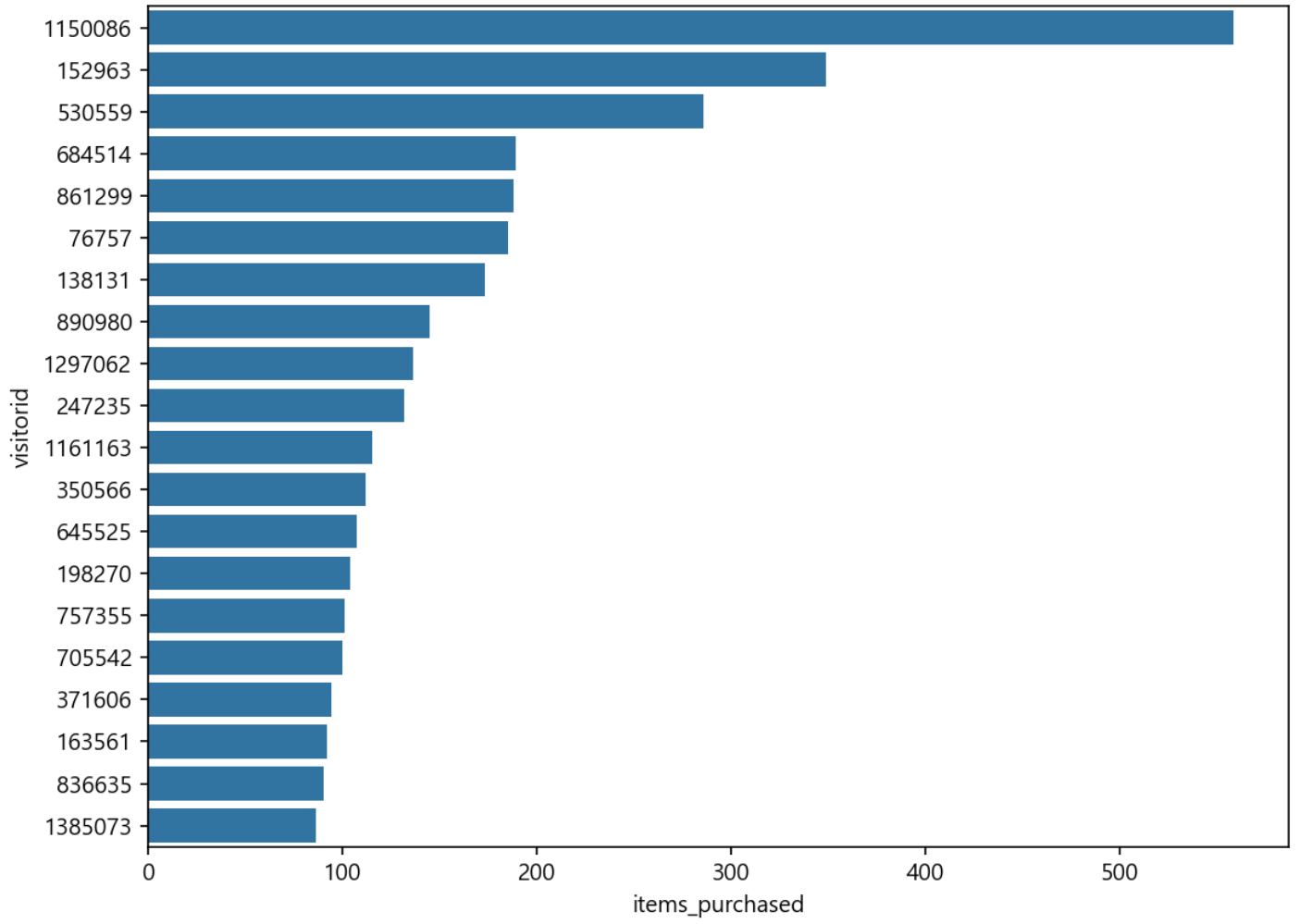
count 11719.000000 11719.000000 mean 1.507978 1.916290 std 7.344491 8.850529 min 1.000000 1.000000 25% 1.000000 1.000000 50% 1.000000 1.000000 75% 1.000000 1.000000 max 502.000000 559.000000 ``

████ █ 20 █ (visitorid, transactions, items_purchased)

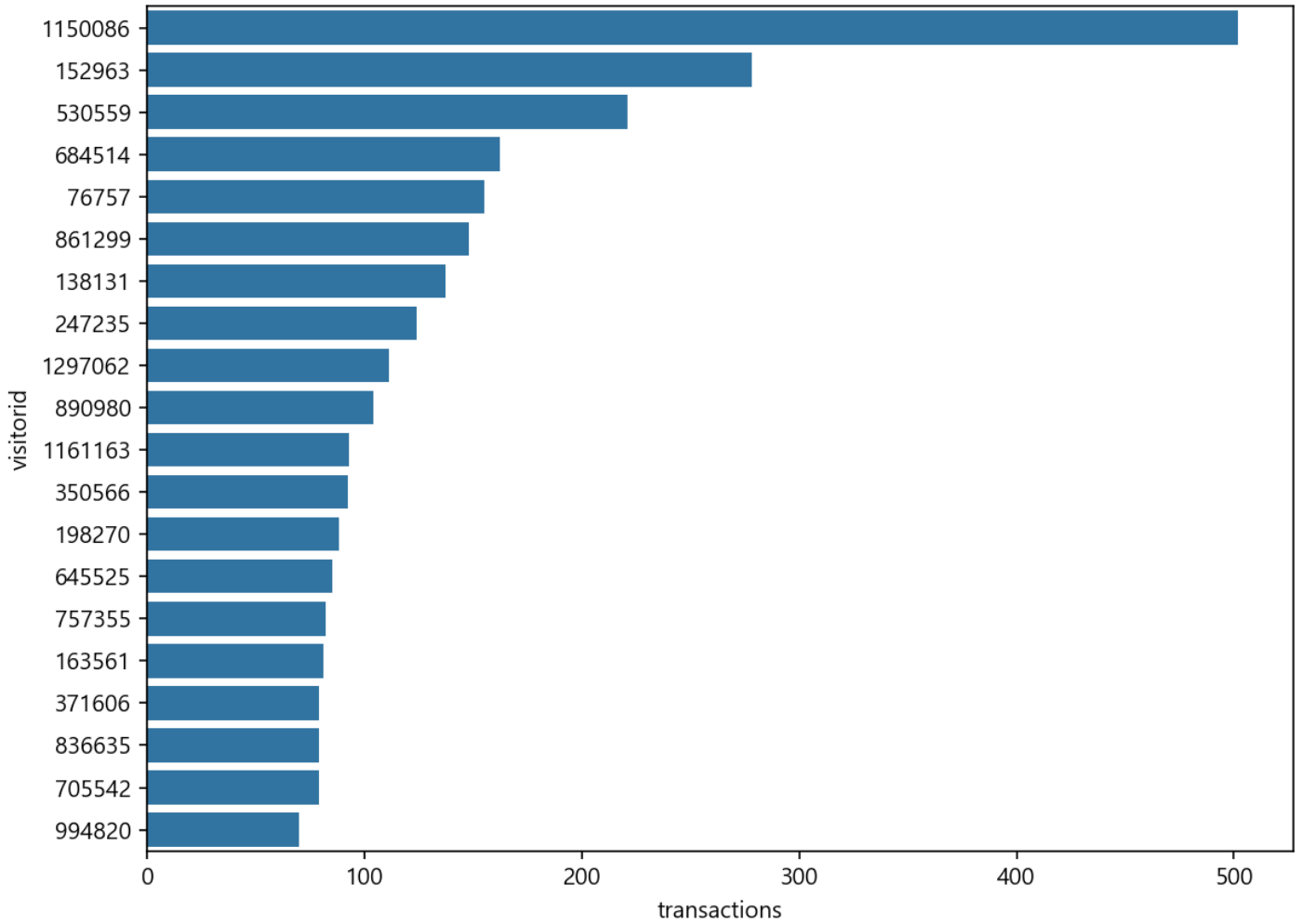
visitorid transactions items_purchased 1150086 502 559 152963 278 349 530559 221 286 684514 162 189 861299 148 188 76757 155 185 138131 137 173 890980 104 145 1297062 111 136 247235 124 132 1161163 93 115 350566 92 112 645525 85 107 198270 88 104 757355 82 101 705542 79 100 371606 79 94 163561 81 92 836635 79 90 1385073 61 86 `` ## ███ █ 20 █ (visitorid, transactions, items_purchased)

visitorid transactions items_purchased 1150086 502 559 152963 278 349 530559 221 286 684514 162 189 76757 155 185 861299 148 188 138131 137 173 247235 124 132 1297062 111 136 890980 104 145 1161163 93 115 350566 92 112 198270 88 104 645525 85 107 757355 82 101 163561 81 92 371606 79 94 836635 79 90 705542 79 100 994820 70 71 ``

구매수량 상위 20 고객



거래수 상위 20 고객



customerid visitorid total_items avg_items segment_items

` customers total_items avg_items segment_items (0.999, 559.0] 11719 22457 1.91629

customerid visitorid

- `SELECT customerid, visitorid, SUM(amount) AS total_items, AVG(amount) AS avg_items, COUNT(DISTINCT segment_items) AS segment_items FROM orders WHERE customerid = '1150086' AND visitorid = '1150086' GROUP BY customerid, visitorid`
- `SELECT customerid, visitorid, SUM(amount) AS total_items, AVG(amount) AS avg_items, COUNT(DISTINCT segment_items) AS segment_items FROM orders WHERE customerid = '1150086' AND visitorid = '1150086' GROUP BY customerid, visitorid`
- `SELECT customerid, visitorid, SUM(amount) AS total_items, AVG(amount) AS avg_items, COUNT(DISTINCT segment_items) AS segment_items FROM orders WHERE customerid = '1150086' AND visitorid = '1150086' GROUP BY customerid, visitorid`